

ELKINS RANDOLPH COUNTY AP, WV, USA

WMO: 724170

Lat: 38.890N

Lon: 79.855W

Elev: 598

StdP: 94.35

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13729

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-13.7	-21.3	0.6	-14.7	-18.3	0.8	-11.8	9.8	3.0	8.7	2.4	1.0	270	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	30.2	21.4	29.1	20.8	28.0	20.3	22.9	28.1	22.2	27.1	21.5	26.2	2.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	17.1	25.3	20.6	16.4	24.6	20.0	15.8	24.0	70.4	28.0	67.7	27.2	65.2	26.1	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.1	7.1	5.8	-22.8	32.4	3.3	1.5	-25.1	33.5	-27.0	34.4	-28.8	35.3	-31.2	36.4
			-22.6	24.3	2.9	0.9	-24.7	24.9	-26.4	25.5	-28.0	26.0	-30.1	26.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-1.4	0.4	4.5	10.6	15.2	19.4	21.3	20.9	17.4	11.2	5.3	1.2
	DBStd	9.35	7.08	6.36	5.89	5.04	4.25	2.98	2.51	2.43	3.57	4.71	5.09	5.73
	HDD10.0	1357	358	273	187	54	8	0	0	0	1	43	157	276
	HDD18.3	3129	611	502	427	235	113	24	6	6	58	225	392	530
	CDD10.0	1561	5	4	18	71	171	283	349	337	224	80	15	4
	CDD18.3	292	0	0	0	2	17	57	97	85	31	4	0	0
	CDH23.3	2551	0	1	8	75	223	489	751	653	296	53	2	0
	CDH26.7	574	0	0	1	10	34	99	198	162	64	7	0	0
Wind	WSAvg	1.9	2.5	2.6	2.6	2.6	1.9	1.6	1.3	1.2	1.3	1.7	2.0	2.2
Precipitation	PrecAvg	1127	78	75	102	97	109	110	122	104	98	73	82	87
	PrecMax	1492	159	143	225	176	195	255	223	264	226	160	281	171
	PrecMin	865	27	20	37	26	37	42	33	28	8	8	22	23
	PrecStd	144	33	28	43	39	48	51	49	47	44	38	46	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.3	20.1	24.3	28.1	29.3	30.6	31.9	31.3	30.5	27.4	22.6	18.8
		MCWB	13.4	13.3	14.1	15.4	19.3	21.2	22.7	22.3	19.5	18.1	14.2	13.7
	2%	DB	15.8	16.2	21.0	25.5	27.6	29.1	30.2	29.9	28.6	24.7	19.5	16.0
		MCWB	11.8	10.4	12.6	15.0	18.2	20.8	22.1	21.7	19.6	16.7	12.6	11.9
	5%	DB	12.8	13.4	17.9	23.3	26.0	27.8	28.9	28.5	26.8	22.7	17.1	13.4
		MCWB	9.8	9.3	11.0	13.9	17.7	20.4	21.3	21.1	19.2	15.7	11.7	10.1
	10%	DB	9.5	10.7	14.9	20.8	24.1	26.4	27.6	27.1	24.8	20.3	14.7	10.8
		MCWB	6.7	7.0	9.8	12.6	16.9	19.7	20.8	20.6	18.3	14.7	10.4	7.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	14.2	15.2	17.7	20.9	22.9	24.0	23.9	22.3	20.2	16.0	14.6
		MCDB	17.4	18.2	22.3	24.6	26.9	28.2	29.6	29.2	27.5	25.3	20.5	17.8
	2%	WB	12.3	11.8	13.6	16.3	19.9	22.0	23.0	22.8	21.1	18.6	14.2	12.6
		MCDB	15.2	14.9	18.9	22.8	25.5	26.8	28.4	27.8	25.9	22.6	18.0	15.2
	5%	WB	10.2	9.8	12.3	15.2	18.9	21.2	22.3	22.0	20.2	16.9	12.7	10.7
		MCDB	12.6	13.0	16.5	20.9	23.9	26.3	27.2	26.7	24.6	20.8	15.7	13.0
	10%	WB	7.1	7.3	10.5	14.0	17.8	20.5	21.6	21.3	19.3	15.4	11.0	8.3
		MCDB	9.4	10.5	14.5	19.4	22.5	25.3	26.1	25.6	23.2	19.2	14.2	10.3
Mean Daily Temperature Range	5% DB	MDBR	10.7	11.5	12.6	14.5	13.1	12.1	11.7	11.8	12.6	13.6	12.7	10.6
		MCDBR	13.3	16.2	18.3	19.1	16.6	14.2	13.4	13.5	15.5	17.4	17.6	14.6
		MCWBR	9.7	11.2	10.5	9.4	8.1	7.0	6.1	6.2	7.5	9.3	11.4	10.7
	5% WB	MCDBR	11.7	14.0	14.9	14.5	13.3	12.1	11.6	11.4	12.3	13.4	13.5	12.6
		MCWBR	9.2	10.6	9.6	8.0	7.1	6.2	5.7	5.6	6.3	8.0	9.7	10.1
Clear-Sky Solar Irradiance	taub	0.292	0.312	0.336	0.377	0.417	0.445	0.464	0.477	0.382	0.346	0.310	0.290	
	taud	2.477	2.408	2.412	2.332	2.223	2.175	2.145	2.079	2.341	2.444	2.517	2.518	
	Ebn at Noon	890	914	923	900	865	838	817	795	861	862	862	868	
	Edh at Noon	79	96	107	123	140	147	151	157	114	92	76	71	
All-Sky Solar Radiation	RadAvg	1.80	2.59	3.62	4.73	5.25	5.82	5.60	5.16	4.30	3.14	2.22	1.59	
	RadStd	0.19	0.38	0.26	0.39	0.45	0.39	0.34	0.27	0.50	0.38	0.25	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.57	N/A	N/A	N/A	N/A	N/A	N/A	-179	+175	+67
Regional (31 neighbors)		+0.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+118	+58

Nomenclature: See separate page